

Memory Technology 2021: Trends & Challenges

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Abstract- DRAM and NAND Flash memory demands for server, datacenter, cloud, mobile and AIOT including ML, autonomous and connected vehicles application have been increased on and on. In addition, emerging memory market such as STT-MRAM, ReRAM, PCRAM, FeRAM and 3D XPoint Memory are expected to reach \$36B by 2030. Memory technologies have quite different stories from Logic devices which is referring to Moore's Law. DRAM, NAND Flash and Emerging memory makers are racing of device scaling, however many of electrical and physical limits are coming into reality. We overview current memory technology, and further discuss the details, trends, and upcoming challenges.

Keywords- Memory, DRAM, NAND, Emerging, MRAM, ReRAM, FeRAM, XPoint, Trends, Challenges

I. INTRODUCTION

DRAM cell scaling down to sub-15 nm design rule (D/R) has already been productized from major DRAM players such as Samsung, Micron, and SK Hynix. They've been developing n+1 and n+2 generations so called 1b (or 1 β) and 1c (or 1 γ), which means DRAM cell D/R might be able to further scale down to sub-12 nm or beyond with/without EUV adoption for DRAM cell patterning [1]. The cell scaling is getting slower due to many challenges such as process integrity, cost, cell leakage, capacitance, refresh management and sensing margin. Some innovative technologies such as higher-k dielectric materials, pillar (or quasi-pillar) capacitor process, recess channel S/A transistor, and HKMG adoption can be seen from the advanced DRAM cell design. Further, 3D DRAM, high bandwidth memory (HBM3), graphic DRAM (GDDR6X/7) and embedded DRAM (10 nm/7 nm) technologies will extend the DRAM lifetime and application [2].

Major NAND manufacturers are racing to increase the number of vertical 3D NAND gates, they all have introduced 1yyL 3D NAND devices, for example, Samsung V7 V-

NAND, KIOXIA and Western Digital Company (WDC) BiCS6, Micron 2nd gen. CTF CuA, and SK Hynix 2nd gen. 4D PUC NAND. Beyond storage density, 3D NAND prototype is used for ultra-low latency NAND application (classified as Storage Class Memory), such as Samsung Z-SSD and KIOXIA XL-FLASH. 3D NAND bit density reached up to 10.8Gb/mm² (SK Hynix 176L 512Gb TLC) and 12.8Gb/mm² (Intel 144L 3-deck QLC) [3]. YMTC 128L Xtacking products are just around the corner.

Intel extends XPoint memory application not only for conventional SSD but also DCPMM persistent memory. Intel Optane™ P5800X SSD products use the 2nd generation XPoint memory technology with four-stack PCM/OTS cell structure. Everspin 3rd generation standalone 256 Mb STT-MRAM (pMTJ) and 1Gb STT-MRAM, Samsung and SONY's new 28 nm eSTT MRAM (pMTJ), Avalanche eSTT MRAM (pMTJ) with 40nm node, Dialog Semiconductor (old Adesto Technologies) 2nd generation CBRAM, and Fujitsu 45 nm ReRAM 130nm FeRAM products have been released on the market in 2020 and 2021 [4, 5].

II. DRAM TECHNOLOGY TRENDS & CHALLENGES

Fig. 1 shows a DRAM roadmap from major and minor players, including Samsung, Micron, SK Hynix, Nanya, PSMC, and CXMT. Three major players, Samsung, Micron, and SK Hynix already revealed D1z and D1a products which have 15 nm and 14 nm-class cell design rule (D/R) for DDR4, DDR5 and LPDDR5 application. Samsung adopted EUV lithography into D1x DDR4 test vehicle (TV) products and D1z LPDDR5 mass products, while Micron and SK Hynix keep ArF-i based double patterning technology (DPT) process for D1z generation. A few more DRAM generation with further scaled down design such as 1d (or 1 δ), 0a (or 0 α), and 0b (or 0 β) will be productized by 2030. Another DRAM maker, CXMT, from China jumped into the race, and is developing D1x and D1y through this year.

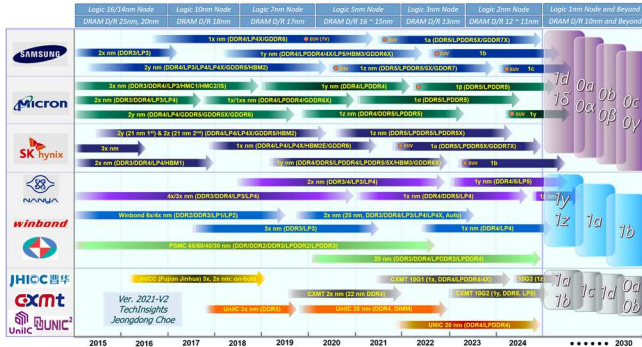


Fig. 1. A DRAM roadmap prepared from TechInsights showing D1z and D1a DRAM products commercialized on the market in 2020 and 2021. A few more generations such as 1d (or 1δ), 0a (or 0α), and 0b (or 0β) will be productized by 2030.

Up to date we've seen $8F^2$ and $6F^2$ DRAM cell design, in which unit cell comprises of 1T (transistor) plus 1C (capacitor). The 1T+1C cell design will be used on DRAM cell design for next decade, however due to the process and layout limitation, DRAM players have been developing $4F^2$ cell structure such as 1T DRAM or capacitorless DRAM prototypes for one of the next candidates to extend DRAM technology (Fig. 2).

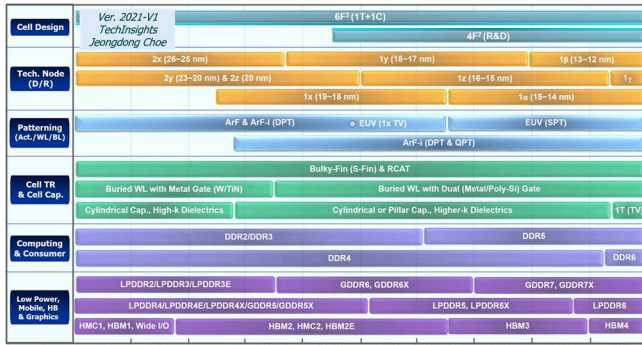


Fig. 2. A technology/application roadmap on DRAM devices showing a $6F^2$ 1T+1C cell design extended to a few more next DRAM generations, although DRAM players have been developing $4F^2$ cell structure such as 1T DRAM or capacitorless DRAM prototypes.

Bulky fin (or saddle fin) with B-RCAT structure for cell access transistor has been used, however the buried wordline gate materials already changed from single tungsten layer to poly-Si/tungsten double layer to control gate leakage effectively. Further, Micron uses TiN-only gate for D1z cell integration. A cylinder-type structure was the mainstream for DRAM cell capacitor integration, but SK Hynix (D1y and D1z) and Samsung (D1z) adopted a quasi-pillar capacitor structure in which cell capacitor uses the outer surface of the cylinder only, which causes smaller cell capacitance than previous generation. In a couple of years, DDR5, GDDR7,

LPDDR6 and HBM3 products would be common on the market.

For 10 nm-class and beyond DRAM cell design, more innovative process, materials, and circuit technologies should be added into them, including higher-NA EUV, $4F^2$, 1T DRAM, pillar capacitor, ultra-thin higher-k capacitor dielectrics, and low-k ILD/IMD materials (Fig. 3).

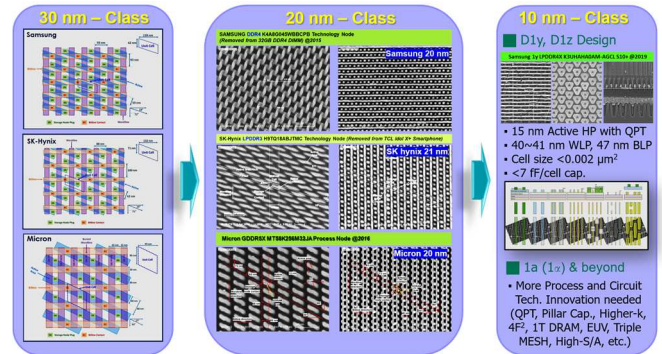


Fig. 3. DRAM Cell design and technology trends from 30 nm-class through 10 nm-class. More innovative technologies are needed to meet cell capacitance, scaling and speed improvement.

Fig. 4 shows a DRAM design rule (D/R) trend for major players. Given that they keep $6F^2$ DRAM cell design with 1T+1C structure, 10 nm D/R would be the last one for DRAM in 2027 or 2028. Lots of challenges would be there on DRAM scaling such as 3D DRAM, row hammer scaling (circuit), low power design, refresh time scaling and management, low latency, new work-function materials, HKMG transistor, and on-die ECC. The most keywords would be 'speed' and 'sensing margin'. Samsung's HKMG peripheral transistor for DDR5 is an example to increase BL sensing margin.

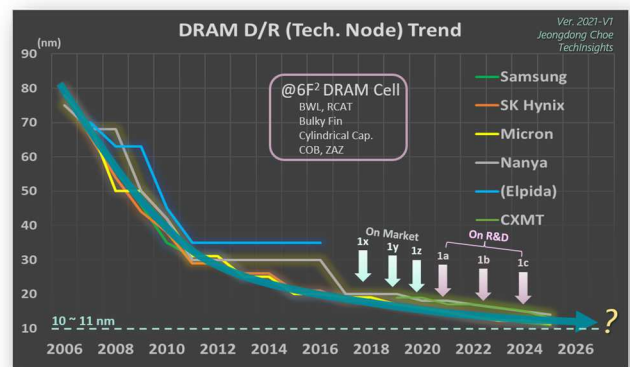


Fig. 4. DRAM D/R trends showing a limitation of $6F^2$ cell design. 10 nm D/R would be the last one for $6F^2$ DRAM in 2027 or 2028.

III. NAND TECHNOLOGY TRENDS & CHALLENGES

Major NAND chip manufacturers are racing to increase the number of vertical 3D NAND gates. They have already introduced 1yyL 3D NAND devices up to date. Samsung 176L (V7), KIOXIA/WD 162L (BiCS6), Micron 176L (2nd CTF), and SK Hynix 176L (V7) are for 1yyL products, and YMTC 128L Xtacking will be introduced on the market in 2021 (Fig. 5). MXIC also announced their first 3D NAND prototype with 48L and will mass-produce by late 2021 or early 2022.

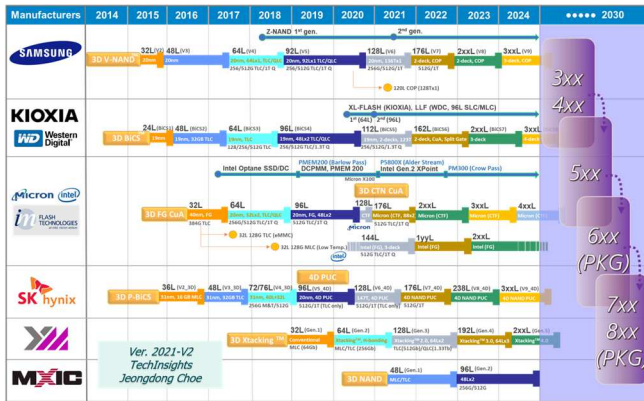


Fig. 5. A 3D NAND roadmap from TechInsights showing 112L/128L and 162L/176L products released on the market in 2021. Z-NAND, XL-FLASH, and XPoint are added into the roadmap for SCM or fast-NAND application.

Some innovative technologies and designs have been adopted so far, such as triple-deck structure, CuA/COP/PUC technology, and Xtacking dies with H-bonding. Further, low latency (high speed) NAND products such as Samsung Z-NAND and KIOXIA XL-FLASH have successfully commercialized with 3D NAND cell architecture and multi-plane die design. For more than 500-layer NAND products, we should consider not only multi-stack or die-stack integration but also 3D package solution.

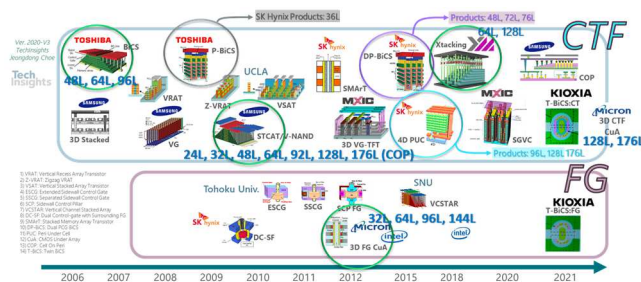


Fig. 6. Seven different 3D NAND prototypes have been proposed and successfully productized: V-NAND, BiCS, FG CuA, CTF CuA, P-BiCS, 4D PUC, and Xtacking.

Since 2018, most of the worldwide smart phones have used 3D NAND storage components instead of 2D NAND chips. Up to date, seven different 3D NAND prototypes have been

proposed and productized; V-NAND from Samsung, BiCS from KIOXIA (old Toshiba Memory) and Western Digital, FG CuA from Intel/Micron, CTF CuA from Micron (128L~), P-BiCS from SK Hynix (~72/76L), 4D PUC from SK Hynix (96L~), and Xtacking from YMTC (Fig. 6 and Fig. 7).

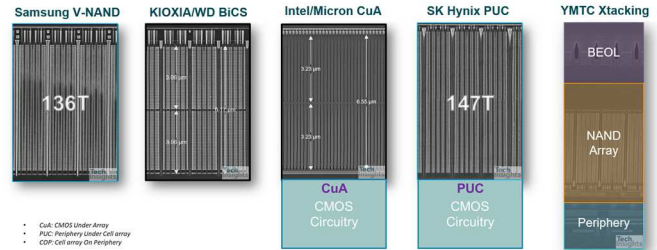


Fig. 7. Five representative SEM images showing the concepts of each 3D NAND cell array architecture. CuA, PUC and Xtacking prototypes have the CMOS peripheral circuitry under the NAND cell array.

Especially, Samsung V-NAND (TCAT) 3D NAND products applied a single VC etching process up to 128L (V6), while all the other 3D NAND products use multi-deck (three-deck for Intel 144L, for example) string integration. They all use 20 nm or 19 nm BL half-pitch, which means ArF-i and DPT based photolithography is the major patterning technology for 3D NAND.

2D NAND wafers are still needed for some specific application with higher reliability and low/high temperature operation. MCU, medical, robots, TVs/toys, game controllers, wearables, security cameras, smart speakers, IoT, AI, ML, printers, set-top box, and aerospace need 2D NAND products. Now, 3D NAND products are very popular for datacenter, cloud, server, SSD, PCs, mobile, and smart phones. As the number of gates stacked increases, the vertical NAND string height is also increased, for example, newly minted 176L products show a 12 μm height from source plate (Fig. 8). Bit cost decreased on and on, and bit density increased up to 15Gb/mm² for QLC die. Total number of gates per NAND string also increased to 200 gates or more.

Intel 144-tier NAND string consists three decks (upper deck, middle deck, lower deck and 48L for each) between source and bitline for the first time and keeps FG CuA structure for TLC and QLC devices. Each deck can be assigned to any combination of QLC or SLC blocks to fully benefit from Intel's new block-by-deck concept in the storage system.

We cannot yet know all the detailed challenges for future 3D NAND technology, however some of them are HAR, layer stress control, wafer warpage, process uniformity, strict control of ALD/ALE, throughput, deck-to-deck misalign, yield control, defects, NAND string current, decoder TR reliability, PGM/ERS speed, retention, e-migration, leakage

and interference, 3D package solution, and so on. PLC 3D NAND products may will be introduced in a couple years.

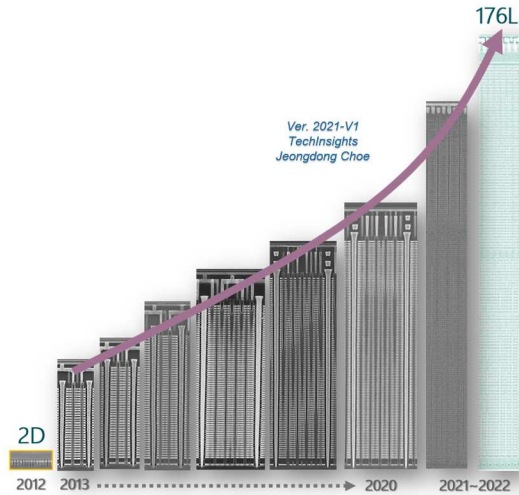


Fig. 8. A comparison of 3D vertical NAND string heights. Newly minted 176L products show a 12 μm height from source plate.

IV. EMERGING MEMORY TECHNOLOGY TRENDS & CHALLENGES

For a couple of decades, we've considered MRAM (or STT-MRAM), PCRAM, ReRAM and FeRAM devices and technologies as emerging memory prototypes. However, they'll be rather a kind of nonvolatile memory devices for embedded application than a discrete emerging memory device. Future emerging memory might be referred to novel magnetic memory devices such as SOT MRAM, FTJ, unipolar or bipolar filamentary OxRAM, CBRAM, macromolecular memory, Mott memory or DNA storage. Here, we still consider the MRAM, XPoint, ReRAM (CBRAM) and FeRAM for emerging memory devices. They're extending application area such as CPU/APU cache memory (STT-MRAM), AI and in-memory computing (PCRAM), analog IC (ReRAM, Memristor), external switch (FeRAM), and high density SCM (XPoint Memory).

Among the emerging memory devices, especially STT-MRAM technology has been actively researched and developed by major players/developers such as Everspin Technologies, GlobalFoundries, Avalanche Technologies, Sony, Micron, IMEC, CEA-LETI, Applied Materials, Samsung, Fujitsu, IBM, TSMC, and Spin Transfer Technologies (STT). Intel, Micron, and SK Hynix are the developers more focusing on XPoint memory with PCM/OTS cell structure. Micron just exited from XPoint memory in 2021 (Fig. 9).

Up to date we've found Everspin 3rd generation standalone 256Mb STT-MRAM (pMTJ) and 1Gb STT-MRAM,

Samsung and SONY's 28 nm eSTT MRAM (pMTJ), Avalanche eSTT MRAM (pMTJ) with 40 nm node, and Dialog Semiconductor (old Adesto Technologies) 2nd generation ReRAM (CBRAM) products from the market. TSMC announced a 22 nm eMRAM-F product roadmap to replace eFLASH for data/code storage and configuration memory application. They're currently developing eMRAM-S, and 14 nm/12 nm eMRAM-F/eRRAM as well.

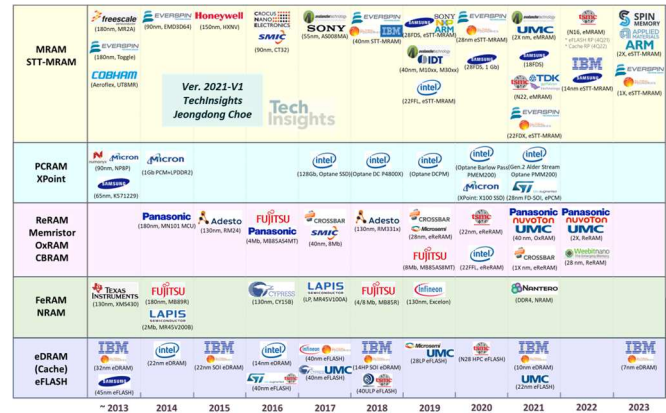


Fig. 9. An Emerging memory roadmap from TechInsights including STT-MRAM, PCRAM/XPoint, ReRAM/CBRAM, FeRAM and embedded DRAM/FLASH memory.

Everspin has four different MRAM products released on the market, including Toggle-mode MRAM (1st generation, Chandler fab.) and STT-MRAM (2nd ~ 4th generation, GF fab.). Among the STT-MRAM products, the 2nd generation STT-MRAM devices use a MgO-based in-plane MTJ structure, while the 3rd and 4th generation STT-MRAM devices use a perpendicular MTJ (pMTJ) technology. Avalanche pMTJ STT-MRAM cell design and structure show 40 nm p-MTJ layers with 0.032 μm^2 cell size, and MRAM layer placed under M1 source line, between Contact-1 and Contact-2. Samsung, together with Sony, revealed 28 nm pMTJ 8Mb embedded STT-MRAM structure for GPS controller of Huawei GT2 smartwatch, for example. They're MgO MTJ-based devices.

Fujitsu 8Mb ReRAM device is the world's largest density as a stand-alone mass-produced ReRAM product up to date. Fujitsu adopted a new 45 nm CMOS process with quite smaller die size and higher memory density compared with their previous 180 nm 4Mb ReRAM products.

The 1st generation XPoint memory die from Intel and Micron has 128Gb (16GB) die density and two-layer PCM/OTS structure. It has been used many of Intel SSD products such as Optane, 800P, 900P, DC P4800X, H10/H20 and DCPMM. For the storage elements, many candidates such as phase change materials, resistive oxide cells, conductive bridge cells and MRAM cells have been proposed and developed.

Among them, the 1st gen. XPoint memory adopted a chalcogenide phase change material, GST (Ge-Sb-Te) alloy layer.

A 20 nm double patterning technology (DPT) used for BL and WL lithography/etching process, effectively 2F² cell designed. Recently, Intel released the 2nd generation XPoint memory, for example, Optane DC P5800X SSD products on the market. 4-stack PCM/OTS layer structure, effectively 1F², is integrated on M4 layer, results in WL/BL/WL/BL/WL multi-layers. The Ovonic Threshold Switching Selector (OTS) co-integrated with a PCM layer for the device, which is the same elements with previous XPoint 1st generation (Fig. 10).

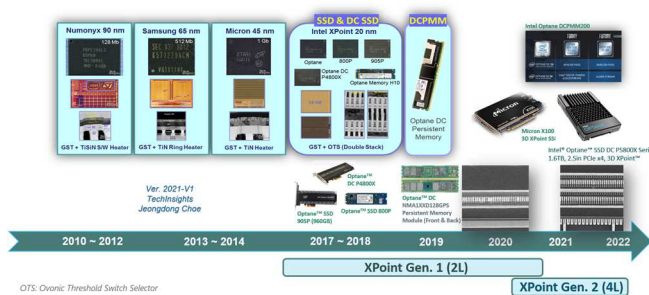


Fig. 10. A PCM/XPoint history showing 2L 1st gen. and 4L 2nd gen. XPoint memory products from Intel.

Although emerging memory devices have opportunity to replace eFLASH or SCM due to high performance (high speed, endurance, and retention) and energy efficiency. However, one of the most challenges would be how to reduce the bit cost, in other words, how to increase the array cell density. So far, all the standalone STT-MRAM dies (256Mb or 1Gb) and XPoint dies (128Gb or 256Gb) are not comparable with 3D NAND dies (1Tb or 1.33Tb for QLC NAND dies). In addition, most of the emerging memory devices use some new materials such as HfO, HZO, GST-based chalcogenide compound, and Ir/Ta-based metal electrodes which causes some difficulties on process integration including patterning/etching, deposition, and annealing optimization.

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